AI-POWERED LIFESTYLE ASSESSMENT

Multilingual Clinical Dialogue System

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Django · Next.js · Groq · scispaCy · ChromaDB

Why Is This Feature Needed?

THE PROBLEM

- Many diseases are strongly influenced by lifestyle habits
- Doctors have limited consultation time
- Patients may forget to mention lifestyle habits
- Language barriers make detailed history-taking harder
- Lifestyle information is often incomplete
- Important health risks may go unnoticed

THE SOLUTION

An AI assistant collects lifestyle information during the doctor–patient conversation in the patient's own language, scores it against a fixed clinical rubric, and gives the doctor a structured summary with guideline-backed recommendations — which the doctor can review and override before anything counts.

Objective

1. Collect lifestyle information during clinical conversations
2. Analyse patient habits across 8 lifestyle areas
3. Detect possible lifestyle-related health risks
4. Generate a deterministic Lifestyle Wellness Score
5. Provide guideline-grounded personalised recommendations
6. Give doctors a reviewable, editable clinical summary

System Workflow

1

Patient Speaks

“I've been feeling tired for two weeks.”
Voice or text, 9 languages

2

AI Asks Questions

One lifestyle area at a time, in the patient's own language

3

Patient Answers

Extracted into structured fields with a source quote

4

System Analyses

Score and risk flags computed in plain Python

5

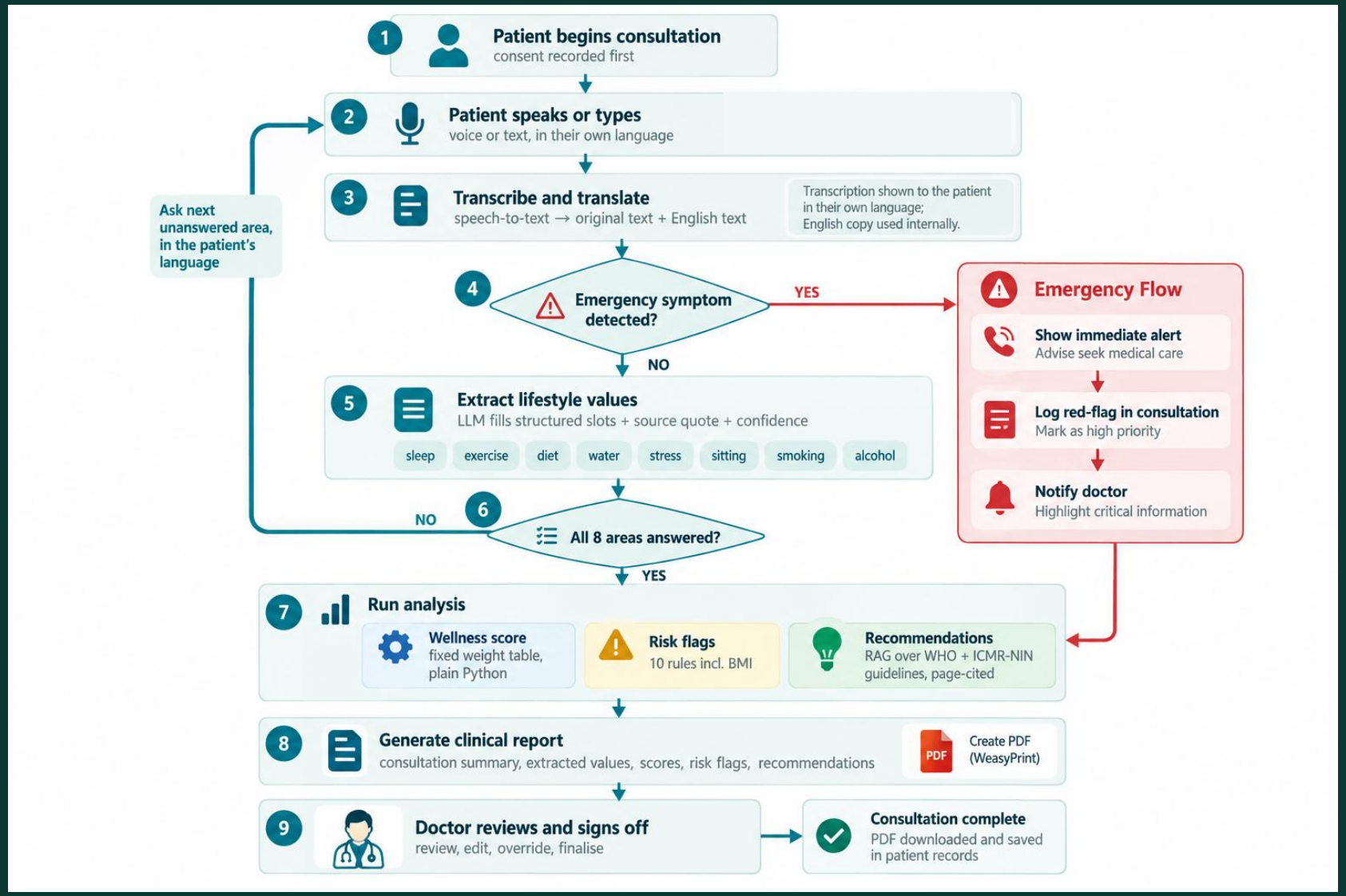
Doctor Reviews

Summary, score, risks, recommendations, PDF

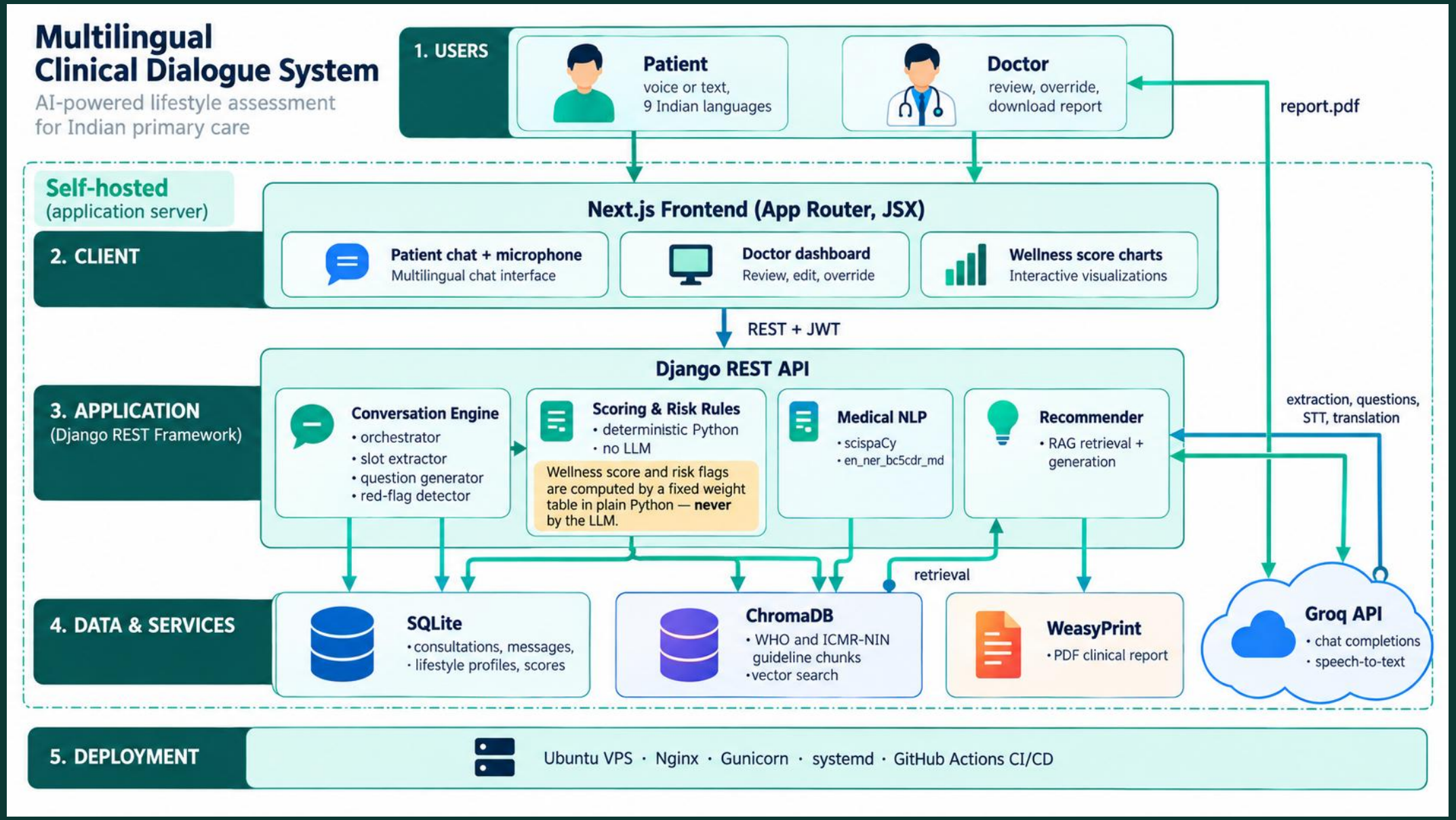
Emergency override

If the patient reports chest pain, breathlessness, stroke signs or similar, the questionnaire stops immediately, the consultation is flagged Escalated, and the patient is told to seek in-person care now.

System Workflow



System Architecture



Lifestyle Assessment Questions

The AI asks these one at a time, adapts to what the patient already said, and moves on after one failed follow-up rather than pushing.

?

Hours of sleep?

?

How often do you exercise?

?

Do you smoke?

?

Do you drink alcohol?

?

What does your diet look like?

?

Water intake daily?

?

Stress level?

?

Long hours sitting?

AI Analysis & Example Output

WHAT THE SYSTEM ANALYSES

- Extracts structured lifestyle values from free-form speech
- Runs medical NER over the transcript for symptoms
- Calculates a Wellness Score from a fixed weight table
- Flags lifestyle-related risks with a stated reason
- Retrieves matching guideline text for recommendations

EXAMPLE RISK DETECTION

Poor sleep · High stress · Sedentary lifestyle · Unhealthy diet · Low water intake · Obesity (BMI) · Possible hypertension risk

CLINICAL SUMMARY (REAL OUTPUT)

Symptoms: Fatigue (2 weeks), mild headaches, no fever

Sleep	5 hrs/day
Exercise	None
Diet	Frequent fast food
Water Intake	1 litre/day
Stress	High
Sitting	9 hrs/day
Smoking	No
Alcohol	Occasionally

Wellness Score **34 / 100 — Poor**

Health Risk Factors & AI Recommendations

RISK FACTORS DETECTED

Poor sleep — 5 hrs/day, below the 6-hour threshold

Sedentary lifestyle — no exercise, 9 hrs sitting

High stress

Unhealthy diet — frequent fast food

Low water intake — 1 L/day

Obese (BMI) — BMI 27.0, at or above the Asian-Indian cutoff of 25.0

Possible hypertension risk — 3 contributing factors present

AI RECOMMENDATIONS

Short bouts of yoga or pranayama, 5–10 min several times a day

ICMR-NIN Dietary Guidelines

Build up to 150 minutes of moderate activity per week

WHO Physical Activity Guidelines, p.42

Replace fast food with fruits, vegetables and whole grains

ICMR-NIN Dietary Guidelines, p.88

Increase water intake to 2–3 litres a day

Aim for 7–8 hours of sleep with a consistent bedtime

Every recommendation is retrieved from an actual guideline PDF and cites its page. If nothing relevant is found, the system says so and falls back to general guidance rather than inventing a citation.

AI & NLP Technologies Used

Speech Processing

Speech Processing
Groq Speech-to-Text API
transcription and translation

Conversational AI

Groq · openai/gpt-oss-120b
extraction and question generation

Medical NLP

scispaCy · en_ner_bc5cdr_md
symptom and disease NER

NLP Processing

Named Entity Recognition
+ regex unit parsing

Knowledge Retrieval

ChromaDB · RAG over WHO and
ICMR-NIN guideline PDFs

Backend & Database

Django + DRF · SQLite
Next.js frontend

How the Wellness Score Is Calculated

Deterministic, not AI-generated

The AI only fills in values. The score itself is a fixed weight table in plain Python — the same inputs always give the same score, and it can be audited line by line.

Factor	Weight	Full marks when
Sleep	15	7–9 hrs/day
Exercise	15	≥150 min/week
Diet	15	Balanced
Stress	15	Low
Smoking	15	None
Alcohol	10	None
Sitting	10	Under 6 hrs/day
Water	5	2–3.5 L/day

BANDS

80–100 Good

60–79 Fair

40–59 Needs attention

Below 40 Poor

- Unanswered areas are excluded, not scored as zero — and under 4 of 8 answered, no score is produced at all.
- Both ends are bounded: 5 litres of water a day is flagged for review, not rewarded.

Advantages & Applications

ADVANTAGES

- Supports preventive healthcare
- Saves doctors' consultation time
- Gathers structured lifestyle data automatically
- Works in the patient's own language
- Improves patient awareness
- Helps detect chronic disease risks early
- Improves clinical decision-making

APPLICATIONS

Diabetes Management

Hypertension Monitoring

Obesity Management

Cardiovascular Disease Prevention

Telemedicine Platforms

Rural & Multilingual Hospitals

Safety & Clinical Guardrails

A screening and documentation aid — never a diagnostic tool.

Never diagnoses	No diagnosis, no medication names, no prescriptions
Emergency escalation	Red-flag symptoms stop the questionnaire and redirect to in-person care
Doctor overrides everything	Any extracted value can be corrected; the score recalculates live
Evidence trail	Every value stores the patient's own words and a confidence level
Insufficient data	Under 4 of 8 answers → no score, marked “insufficient responses”
Age-matched advice	An adult never receives advice written for children
Explicit consent	Recorded before any data is collected, with a written disclaimer

What Was Built

Delivered and verified end to end

- Voice and text consultation in languages, end to end
- 8-area lifestyle assessment with slot-by-slot extraction
- Deterministic wellness scoring and 10 risk rules including BMI
- RAG recommendations cited to WHO and ICMR-NIN guidelines
- Doctor dashboard with live re-scoring and value overrides
- Doctor-led in-clinic consultation mode
- PDF clinical report with full Indic script support

CONCLUSION

Transforming clinical conversation into comprehensive health evaluation

By combining clinical symptoms, lifestyle habits, AI-based risk analysis and personalised recommendations, the system supports preventive healthcare, improves clinical decision-making, and promotes healthier long-term patient outcomes.

Thank You